5. Wednesday, April 9. Sets (2.3) and basic set operations (5.1)

Definition 1: Let A and B denote sets. We say that A is a **subset** of B if whenever $x \in A$, we also have $x \in B$. Notation: $A \subseteq B$ (\subseteq)

Remark: For any set A, we have $A \subseteq A$. Indeed: if $x \in A$, then $x \in A$. Example: $\mathbb{N} \subseteq \mathbb{Z}$.

Definition 2: We say that two sets A and B are **equal** if they have exactly the same elements. In other words, if $A \subseteq B$ and $B \subseteq A$. Notation: $A = B$.

- We say that A is **properly contained** in B if $A \subseteq B$ but $A \neq B$. Notation: $A \subsetneq B$ (\subsetneq)

Example: $\mathbb{N} \subsetneq \mathbb{Z}$.

Definition 3: The **empty set** $\emptyset = \{\}$ (\emptyset) is the set with no elements.

Q: $\emptyset \subseteq \{1,2,3\}$? Is it true that if $x \in \emptyset$ then $x \in \{1,2,3\}$?

A: It's true! The hypothesis is false, so the conditional is true.

Remark: $\emptyset$ is a subset of any set.

The roster method has obvious limitations:
For example, the set $[0,1]$ of real numbers between 0 and 1 (inclusive) can't be written like that.

Back to statements: Recall "$5x = 2$" is a sentence but not a statement

Definition 4: A **variable** is a symbol representing an unspecified object that can be chosen from a given set U. The set U is called the **Universal set for the variable**.
A **constant** is a specific member of the universal set.

Example: "There is $x \in \mathbb{R}$ such that $5x = 2$" is a statement.

Definition 5 An **open sentence** is a sentence $P(x_1, x_2, \ldots, x_n)$ involving variables $x_1, x_2, \ldots, x_n$ with the property that when specific values from the universal set are assigned to the variables, the resulting sentence is a statement (either true of false)

Example: The open sentence $P(x)$: $5x-2=0$ where the universal set is $\mathbb{R}$ becomes true for $x=\frac{2}{5}$ and false for any other x

Definition: The **truth set** of an open sentence $P(x)$ (one variable) is the subset of the universal set containing the elements x for which $P(x)$ is true.

Example: If $P(x)$ is the sentence x^2-2 with universal set $\mathbb{R}$, then the truth set is $\{-\sqrt{2}, \sqrt{2}\}$.

Set builder notation

Given an open sentence $P(x)$ with universal set U, we can form the set

$$\{x \in U \mid P(x) \text{ is True}\}$$

- $\{$: set
- $\mid$: such that
- $P(x)$ is True : truth set

Example: $[0,1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$ Here $U=\mathbb{R}$ and $P(x)$: $0 \leq x \leq 1$.

Set operations (short interlude, 5.1)

Definition 4: Let A and B be subsets of some universal set U.

1) The **union** of A and B is $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$ (cup)
2) The **intersection** of A and B is $A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$ (cap)
3) The **difference** of A and B is $A \setminus B = \{x \in U \mid x \in A \text{ and } x \notin B\}$ (setminus)
4) The **complement** of A is $A^c = \{x \in U \mid x \notin A\}$

Example 5: Let U be a universal set for a variable x. Take open sentences $P(x)$ and $Q(x)$, and denote their truth sets by A and B resp. Then:

1) $P(x) \vee Q(x)$ has truth set $A \cup B$.
2) $P(x) \wedge Q(x)$ has truth set $A \cap B$
3) $P(x) \wedge \neg Q(x)$ has truth set $A \setminus B$ (and $P(x) \wedge \neg Q(x) \equiv \neg(P(x) \to Q(x))$)
4) $\neg P(x)$ has truth set A^c

Example 6: In the universal set $\mathbb{R}$, let $A=[-1,1]$ and $B=[0,2]$. Then

- $A \cup B = [-1,2]$
- $A \cap B = [0,1]$
- $A \setminus B = [-1,0)$
- $A^c = (-\infty,-1) \cup (1,\infty)$

Quantifiers (2.4)

Recall: "$5x+14<2$" is not a statement, but an open sentence (say the universal set is $\mathbb{R}$)
We can make it a statement if we quantify the variable x.

Definition 7: The phrase "There exists" is called an existential quantifier

Notation: $\exists$ \exists

The phrase "For all" is called a universal quantifier

Notation: $\forall$ \forall